

**Example 6.2.** Consider the two-dimensional semilinear parabolic problem

$$\frac{\partial u}{\partial t}(t, x, y) - \frac{\partial^2 u}{\partial x^2}(t, x, y) - \frac{\partial^2 u}{\partial y^2}(t, x, y) = \frac{1}{1 + u(t, x, y)^2} + \frac{1}{1 + u(\frac{t}{2} - \frac{1}{2}, x, y)^2}, \quad (6.63)$$

for  $(x, y) \in [0, 1]^2$  and  $t \in [0, 3]$ , subject to homogeneous Dirichlet boundary conditions. The initial condition is given by  $\phi(t, x, y) = e^{-t}x(1-x)y(1-y)$  for  $t \in [-\frac{1}{2}, 0]$ .

We apply standard finite differences with  $n = 200$  grid points to discretize the problem in each spatial direction. In this example the exact solution is unknown. The reference solution is computed by the ERKC-C method with Gauss collocation points ( $s = 3$ ) using the constant step size  $h = 2^{-11}$ . The errors of the ERKC-I and ERKC-C methods in the  $L^\infty(\Omega)$  and the  $L^2(\Omega)$  norm at final time  $T = 3$  are presented in Figures 3 and 4. The figures confirm the theoretical analysis. Moreover, a comparison of computational cost of the ERKC methods with Gauss collocation points ( $s = 3$ ) is presented in Table 1. For step sizes  $h = 2^{-k}$  with  $k = 3, 4, \dots, 9$ , the ERKC-I method consistently reduces the CPU time by approximately 12% compared to the ERKC-C method, demonstrating its computational efficiency across a range of step sizes.

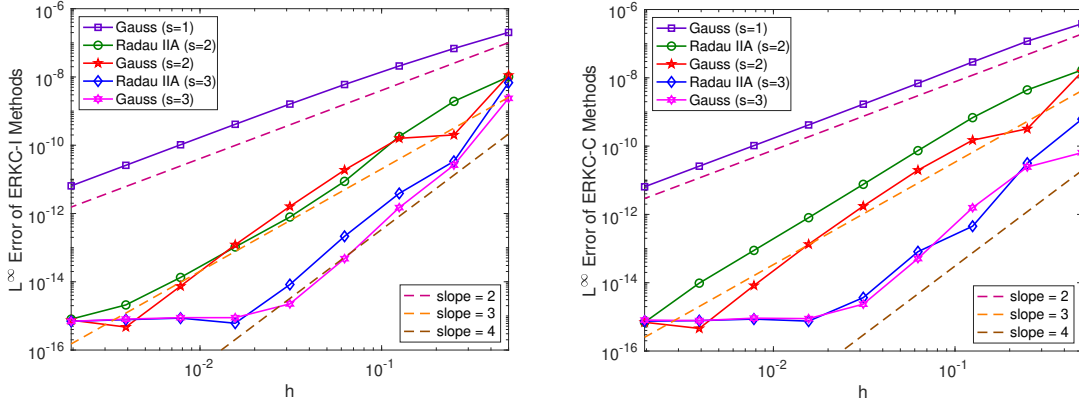


Figure 3: The convergence rates of ERKC-I methods (in the left panel) and ERKC-C methods (in the right panel) for (6.63). The errors are measured at  $T = 3$  in the  $L^\infty(\Omega)$  norm.